

Assignment 02

1. Why we are using 25Mhz clock to drive the display why not some other frequency?
2. What is the frequency of the H-sync and V-sync signals generated?
3. Can the same 25Mhz signal can be mapped to the ball and paddle movements. Explain why.
4. Explain what are different timing sections in the VGA display standard we are using.
5. What are these front porch, back porch and sync signals? Why do a display need these extra signals to display something? Explain the significance of these signals.
6. What happens if I manipulate the timing numbers of these porch and sync signals from the standard value?
Case1 - The total timing is constant (800 in horizontal) but the front porch and the back porch is changing i.e. (front porch increases by 10px and back porch decreases by 10px and vice versa).
Case2 – The total timing is changed for example 810px horizontal and 600px vertical.
Case 3 – Only the sync signals are changed.
7. How many different colours can you display on the screen using the standard which we are using?
8. Explain how four bits are taken to generate different shades of the same colour?